

NumPy Practice Questions

- Q1. Given a list [1, 2, 3, 4, 5], create a NumPy array from it?
- Q2. Create an array of 10 zeros?
- Q3. Generate an array with values from 10 to 50?
- Q4. Given an array `arr = np.array([10, 20, 30, 40, 50])`, extract the first and last elements?
- Q5. Given two arrays, `a = np.array([1, 2, 3])` and `b = np.array([4, 5, 6])`, perform element-wise addition?
- Q6. Given `a = np.array([1, 2])` and `b = np.array([3, 4])`, calculate the dot product of these two vectors?
- Q7. Given an array `a = np.array([10, 20, 30, 40, 50])`, extract elements from index 1 to 3 (inclusive of start, exclusive of end)?
- Q8. Given an array `a = np.array([10, 20, 30, 40, 50])`, reverse the elements in the array using slicing?
- Q9. Replace all elements in `a = np.array([10, 20, 30, 40, 50])` greater than 25 with 0?
- Q10. Replace all odd numbers in the array with -1?
- Q11. Calculate the square root of each element in an array?
- Q12. Given `arr = np.array([3, 5, 1, 2, 4])`, find the minimum and maximum values and their respective indices?
- Q13. Given `arr = np.array([10, 15, 20, 25, 30])`, filter out elements that are greater than 20?
- Q14. Given `arr = np.array([10, 15, 20, 25, 30])`, select only values that are greater than 10 and less than 25?
- Q15. Given `arr = np.array([4, 6, 8, 10, 12])`, find the indices of all elements greater than 8?
- Q16. How can you set a random seed in NumPy to ensure reproducibility?
- Q17. Given two arrays `a = np.array([1, 2, 3])`, `b = np.array([4, 5, 6])`, calculate the mean and standard deviation of the elements in an array?
- Q18. Given `a = np.array([1, 2, 3, 4, 5])` and `b = np.array([4, 5, 6, 7, 8])`, find the intersection and union of a and b.
- Q19. Given `arr = np.array([10, 20, 30, 40, 50])`, find the maximum, minimum and mean of the array?

Q20. Find the index of the maximum and minimum elements of the above mentioned array.